A SEM analysis for understanding factors leading to AI-use among university students and positive/negative outcomes associated with it

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Abstract

This study aims to investigate the effect that perceived benefits and risks have on Artificial Intelligence use (AI-use), and the resulting positive and negative outcomes due to its usage. We are additionally interested in investigating differences in AI-uptake patterns between men and women using multi-group analysis. The study was completed with a perspective on university students, using cross-sectional survey data collected by Correa (2026), focusing on study habits and AI-adoption among students at the Manizales campus of a Colombian public university.

After a literature review, and exploration of the data set at hand, an initial hypothesis was able to be posed. Unfortunately, due to data limitations, a perceived risks factor was unable to be formed in this hypothesis, but perceived benefits were included using "Study Habits" as a proxy, as well as both positive ("Academic Success") and negative ("AI-use Risks") outcomes.

The model was tested with Confirmatory Factor Analysis (CFA) and Structural Equation Modeling (SEM). The final SEM model (using study habits, AI-use, academic success, and AI-use risks as latent variables), showed significant p-values for all indicators, with generally moderate strength standardized loadings.

The results revealed that the most important factor to academic success was study habits. In contrast to one of our initial hypotheses, study habits also showed a significant positive impact on AI-use. We found AI-use significantly increased AI-use risk, which then had a negative impact on academic performance. Admittedly, results from the final model have to be taken with a grain of salt, as the fit measures showed to be barely acceptable.

Finally, For the multi-group analysis, a simpler model was used, with latent variables; study habits, AI-use, and academic success. This comparison failed at configurate invariance, showing good fit measures for men, while showing bad fit measures for women. This indicates that there could be some difference in relationships between study habits, academic successes, and AI-use between genders.

1 Motivation and literature

1.1 Research motivation/objective

There is no doubt, AI-usage is more prominent than ever. In all aspects of life, whether that be in data analysis, creative work, or even just a simple google search on your phone, AI-powered and -generated content seems to be very difficult to avoid. One said field where the implementation of AI is still being figured out, is in academia. For researchers/staff and students, AI is possibly a very powerful tool, but it's not perfect.

Here at Uppsala University, students in different faculties, programs, classes, and even friend groups are likely all using AI to differing extents, and for different purposes. From our perspective as students, we're interested in learning more about how our peers decide to use AI. What are the perceived risks and benefits to students of AI use, and do they impact learning outcomes? We set out on this project with the goal to **develop a SEM model for AI-use that incorporates perceived benefits, risks, and both negative/positive outcomes as a result of usage. It is also of interest to investigate whether this model differs between groups, specifically genders.**

1.2 Literature

We look to the past literature to gather information and frameworks on what latent variables/structures have been investigated within this context before.

In terms of factors leading to AI-use, two frameworks that consistently pop up in the research are Ajzen's (1991) Theory of Planned Behavior (TPB) and Davis' (1989) Technology Acceptance Model (TAM). The TPB suggests that a person's behavioral intention is shaped by their personal attitudes, social norms, and their perceived behavioral control. Where the TPB can be applied to any context, the TAM was specifically designed for explaining technology adoption, where a person's perception of the technology's usefulness (perceived usefulness) and ease-of-use (perceived ease-of-use) predicts behavioral intention.

Zeng et al. (2026) investigate factors leading to AI adoption for university students from South-Western China. To do so, they incorporate the TPB and TAM, but also Bauer's (1960) Perceived Risk Theory (PRT) and a model that is not attributed to any particular author, the Knowledge Attitude Practice (KAP) model. Simply, the PRT says that perceived risks influence behavior, and the KAP model suggests that personal attitudes (as seen in the TPB) are formed by one's knowledge of the matter. Their hypothesized structural model was:

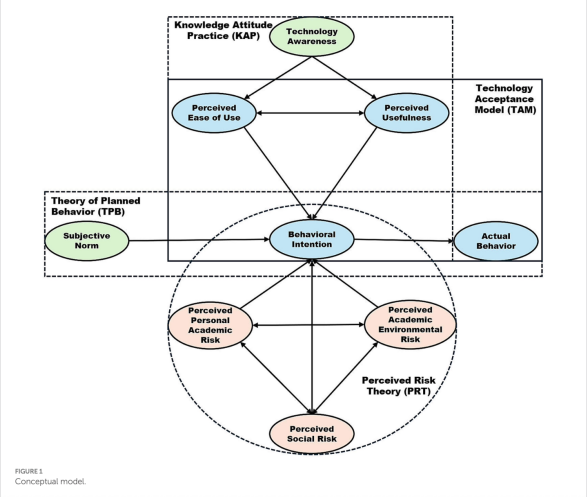


Figure 1: Hypothesis from Zeng et al. (2026)

Of the structural paths seen, they only found one **insignificant** relationship; perceived social risk in determining behavioral intention to use AI.

Ngo et al. (2025) investigated AI-use among university students in Vietnam. They also incorporated the TPB and TAM, as well as perceived trust, a factor that fits within the framework of PRT. They hypothesized that attitude was shaped by perceived benefits and risks, and that attitude in turn affects adoption intention along with subjective norms and perceived behavioral control. All hypothesized paths were found to be significant.

To list out a couple more studies, Sova et al. (2024) found AI-usefulness, training-and-integration, and AI awareness to be either direct or indirectly significant for explaining actual AI-use in Romanian university students. They did not find that concerns about AI significantly affected usage. Saihi et al. (2024) investigate a comprehensive set of 20 factors, of which the TPB is included. Interestingly, they introduce a learning-outcomes factor with a path to the actual adoption-of-AI factor.

Where Saihi et al. (2024) bring up learning-outcomes as a possible explanation for actual AI-use, other studies reverse the arrow. Outcomes "as a result" of AI-use in university settings is also of interest in the literature. Al-Emran et al. (2025) use the TPB and the "Task-Technology Fit" (TTF) model to explain actual AI-use and they in turn found that AI-use significantly and positively affects learning outcomes. Abbas et al. (2024) conducted a study into factors leading to AI-usage and the effect AI-usage has on certain outcomes. They

found significant relationships for that AI-usage was negatively associated with academic performance (contrasting the findings from Al-Emran et al. (2025)), and positively associated with both increased procrastination and memory loss. Hou et al. (2026) ran an experiment and suggested it is *how* students use AI that will determine whether critical thinking skills are lost or not.

When comparing groups in the context of AI-usage in universities, Mohammed et al. (2026) investigated factors leading to AI behavioral intention across universities in Nigeria. They found differences in groups such as gender, degree of studies, and age. Baek et al. (2024) also found differences in AI-usage between the same three groups (among others) at an American college.

Naturally, with the rise in popularity of AI, research into this topic has increased hugely in recent years. We have not covered even a scratch of it, but we certainly have found some significant results that will aid us in factor labeling, indicator selection, and structural path hypotheses.

1.3 Our contribution

As mentioned, we are interested in developing a model that incorporates both positive and negative motivations for using AI, as well as seeing if AI-usage can be attributed to both positive learning and other negative outcomes. This is not something new to the literature, but many SEM studies do appear to focus predominantly on determinants of AI-use, or determinants of AI-use and *either* negative or positive outcomes. We hope to further the research into "dual outcomes", both positive and negative. The second part of our research aim, comparing AI-use patterns between genders, is harder to find (in a clear SEM context) in past research. We believe completing a multi-group analysis and investigating measurement/structural invariance between men and women is possibly an interesting insight into AI-uptake.

2 Data

2.1 Data source and introduction

The data set is a cross-sectional survey focusing on study habits and artificial intelligence use among university students at the Manizales campus of a Colombian public university. The author, Correa (2026), groups the survey questions into six sections as follows; sociodemographic characteristics, 2) study habits, 3) AI tools and use patterns, 4) academic self-perception and satisfaction, 5) ethics and academic integrity, and 6) perceived dependence, creativity, and AI literacy.

With reference to these six sections, at a glance, the data should be directly applicable to studying how AI-use incorporates perceived benefits (*study habits*), risks (*ethics and academic integrity*), learning outcomes (*academic self-perception and satisfaction*), negative outcomes (*perceived dependence, creativity)* and actual AI-usage (*AI tools and use patterns*). In addition, by also having questions related to sociodemographic characteristics like gender, the study is relevant for multi-group analysis.

2.2 Data description

The data was collected during the first academic term of 2025 using a quota based approach, collecting responses until the target sample for each specific program was reached. The sample size contains a total of 357 observations, with the distribution of total students enrolled in different programs at the university kept consistent with the students per study programme in the sample (see table 11 in the appendix for exact breakdown of enrolled students).

Survey responses were completely anonymous, with identifying details, including submission timestamps, removed. The cleaned data set made available by the author, "survey_cleaned_english.csv" provides english variable names, a cleaned GPA variable with a uniform scale, harmonized categorical responses, and a standardized AI tool field with a derived count of the number of used AI tools (Correa, 2026). The author only provides cleaned data sets (one in english, one in spanish) and they both contain no missing values. They do not make it clear at all how they dealt with missing values if there were any.

2.3 Data limitations

Self-reporting-bias
Since the data is entirely self-reported, certain questions are prone to systematic biases. For example, sensitive questions, such as whether a student has failed due to using AI, might not be answered truthfully. Furthermore, questions requiring retrospective estimation, such as hours of study time and risk being under- and over-reported, which is something that should be taken into consideration.

Generalizability
Since the data is only collected from a single public university in Colombia, the dataset might struggle with generalization, both geographically and for different kinds of universities, like private ones, for example, since the data was collected specifically from a public one.

3 Data Exploration

3.1 Response distributions

In the questionnaire, ordinal variables were reported on varying scales. For example, some were reported on a "frequency" scale, containing 5 categories from "never" to "always", whereas others were from "not at all" to "a lot". The "independent study time" variable has a 6 category scale from "0-1" to "5+". There is also a mixture of continuous variables (like GPA) and binary data (like whether the student had failed an assignment due to AI use, "yes" or "no"). For response distribution illustration purposes, ordinal variable category names were replaced with the levels 1 through 5 (or 6). Tables 1, 2, and 3 show the distributions of the ordinal variables, continuous variables, and binary variables, respectively. *Note, some of the binary group variables, like whether the student is a full-time or part-time student, are excluded from this point on as we are only focusing on comparing men and women in this study.*

Looking at Table 1, we see that a majority of students use AI in their studies, with 44.5%, saying they almost always (level 4) use AI, in the *AI-use frequency for study* variable. A clear majority also found AI helpful, with 61.9% saying "Definitely yes" (level 5) to finding AI helpful in the *AI helped understand topics* variable. The students seem divided regarding ethics in the *AI use in academic tasks is fraud* variable, with 37.8% being undecided and 37.5% disagreeing.

Furthermore, the data looks to have a good distribution for the gender variable, making it suitable for multi-group analysis. Looking at Table 3, 58.5% of respondents are male, and 41.5% are women.

Finally, estimator convergence issues can arise when counts in a contingency table are 0. It is impossible to calculate thresholds (as discussed in note 8 by Wallentin (2026)) when you have cells with 0 counts. For this reason, we were happy to see there were no 0's in the ordinal data distribution table (ignoring, of course, the 5 level items having a 0 in the level 6 column). However, some variables do have low values in level 1, which could pose a challenge in the multi-group analysis if some variables then have 0 responses in a given level.

3.2 Correlation Plot

Next, we will look at the correlations of the variables in the data set. When doing this, we exclude our grouped *gender* variable. We calculate the plot below with all relevant correlations as seen in note 8 by Wallentin (2026, Slide 10) with the `lavCor` function from `lavaan` in R.

Looking at the pearson, polychoric, polyserial, tetrachoric, and biserial correlations in Figure 2 below, we are able to gather some insights into our data. After the initial plot, seen on the left of Figure 2, we were able to find what looked like four natural groupings in the data with moderate to stronger correlations. These are portrayed on the right-hand side of the below graph, where we have simply rearranged the ordering and put boxes around the groupings for clarity.

Table 1: Percentage Distribution of Responses for Ordinal Variables

Ordinal Variables	**Level 1**	**Level 2**	**Level 3**	**Level 4**	**Level 5**	**Level 6**
ai_helped_understand_topics	0.6	0.8	4.2	32.5	61.9	0.0
ai_use_frequency_for_study	0.6	7.8	30.5	44.5	16.5	0.0
ai_use_in_academic_tasks_is_fraud	14.6	37.5	37.8	6.7	3.4	0.0
career_satisfaction_affects_motivation	1.7	4.5	20.2	42.0	31.7	0.0
independent_study_time	2.0	5.9	9.5	14.6	16.8	51.3
organizes_notes_before_tests	0.3	4.8	24.6	33.3	37.0	0.0
perceived_ai_dependence	9.5	26.3	26.9	33.9	3.4	0.0
perceived_creativity_reduction_from_ai	13.2	28.6	24.4	28.3	5.6	0.0
procrastination_frequency	0.3	10.6	48.2	33.1	7.8	0.0
program_meets_expectations	1.7	2.5	24.1	47.1	24.6	0.0
prompt_engineering_knowledge	13.2	21.3	26.6	30.0	9.0	0.0
satisfaction_with_grades	1.1	11.8	38.4	44.5	4.2	0.0
self_rated_academic_performance	2.0	18.2	47.9	10.9	21.0	0.0
study_groups_or_tutoring	10.9	36.1	35.3	14.0	3.6	0.0
study_method_effectiveness	0.6	8.1	44.0	41.5	5.9	0.0
uses_supplementary_resources	0.3	6.4	18.2	45.7	29.4	0.0
verifies_work_with_ai_after_completion	9.5	12.6	28.3	32.2	17.4	0.0

Table 2: Descriptive Statistics for Continuous and Numerical Variables ($N = 357$)

Variable	Mean	SD	Median	Min	Max
age_years	20.82	2.76	20.0	17.0	36.0
cumulative_gpa_papa	3.92	0.33	3.9	3.1	4.8
current_program_term	5.78	2.49	5.0	1.0	12.0
n_ai_tools	2.72	1.38	3.0	0.0	11.0

Table 3: Sample Demographic and Categorical Characteristics ($N = 357$)

Variable / Characteristic	n	%
Gender		
Female	148	41.5
Male	209	58.5
Failed Evaluated Activity Due to AI		
No	321	89.9
Yes	36	10.1
Uses AI for Non-academic Activities		
No	33	9.2
Yes	324	90.8

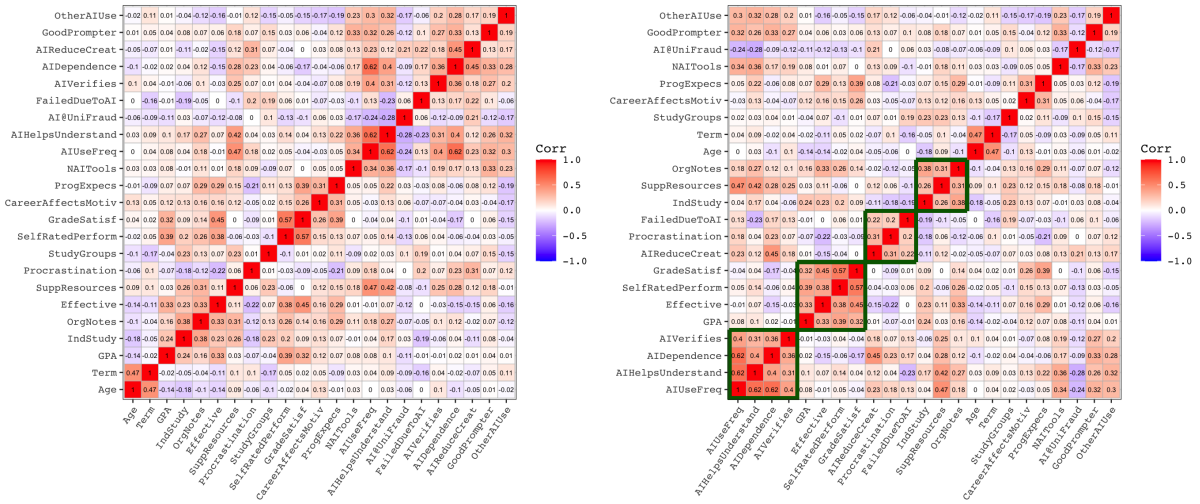


Figure 2: Correlation plot of the variables

Firstly, in the lowest green box, it can be seen how variables reflecting AI-use, such as verifying work, being dependent on it, helping understand topics, and use frequency, show moderate to strong correlations between 0.31 and 0.62. We also see these four variables generally have lower correlations with other variables (except perhaps *uses supplementary resources* and *prompt engineering knowledge*), supporting a potential latent construct from the four AI-use related variables in question.

Moderate-to-strong correlations could also be seen in the second green box relating to academic success, with grade satisfaction, self-rated performance, study effectiveness, and GPA, having correlations ranging from 0.32 to 0.57. These four variables are a more clear grouping than than AI-use, having even lower correlations to other variables in the plot. This might be a good fit for some variable reflecting academic success or performance.

The third box includes some negative aspects, like failing an activity due to the use of AI, and stating that AI reduces creativity. Here, the re-

lationships were not as strong, with the strongest correlation being 0.31 between procrastination and AI reducing creativity. These variables generally correlate weakly with other variables, except reducing creativity with AI dependence.

The final box includes variables that might support a latent variable relating to study habits: organizing notes, using supplementary resources, and independent studying, showing some moderate correlations between 0.26 and 0.38. The highest ones here are supplementary resources and independent studying, with a correlation of 0.38. Supplementary resources is the most suspect variable in this grouping, containing some moderate correlations with the AI-use cluster.

By looking at the correlation plot, some suggestions for latent variables could be supported, these being AI-use, academic success and study habits, with a potential weaker one being AI-use risks. We acknowledge the correlations are not the strongest, and ideally there would be clearer groupings, but this is the best we could see from correlations alone.

3.3 KMO test

The KMO test was performed on all continuous variables and ordered likert scale answers. Nominal categorical variables like gender were excluded since they are not applicable for use with KMO. The KMO test returned a value of 0.709, which is an acceptable result to perform factor analysis on (Fowler, 2025).

4 Analysis Background

4.1 CFA and SEM

Confirmatory factor analysis is based on pre-set theories about latent variables. After identifying possible latent factors from theory, we find observable indicators to reflect these factors in our structured model. The main goal for CFA is then to test these theories and see if they are reflected in data (Jöreskog et al., 2016). For this analysis, the confirmatory factor analysis is shaped by previous CFA examples and research into how AI affects students in academic environments. Structural Equation Modeling will be used to model hypothesized relationships between latent variables.

4.2 Multi-group analysis

We will conduct a multi-group analysis by comparing genders to see how similar their separated models are. We first perform measurement invariance tests. If this performs well, we can move on to structural invariance tests. We follow the same procedure for measurement invariance as seen in note 6 by Wallentin (2026, Slide 14), which tests for: **Configural Invariance** (whether we have the same factor structure across groups), **Metric invariance** (whether the factor loadings are equal across groups), **Scalar Invariance** (whether the indicator intercepts are equal across groups), and finally **Strict Invariance** (which measures if residual errors are equal across groups). A likelihood ratio test (LRT) with the following hypothesis is used to compare each separate fit where loadings, then intercepts, then residuals are progressively constrained to be equal.

$$H_0 = \text{The models are equivalent}$$

$$H_1 = \text{The models aren't equivalent}$$

If there is strong measurement invariance, a test for structural invariance may also be of interest (Wallentin, 2026, Slide 17). The following structural parameters are then tested:

$$\Phi_g = Var(\xi_g)$$

$$E(\xi_g)$$

Here, the constrained loadings, intercepts, and residuals model is then tested with a model with an added constraint on latent variances, through a LRT. Finally, the latter model with constrained variances is tested against the fitted model with an added constraint on latent means, also with the same test. If all of these tests show a high p-value the groups are completely invariant/equivalent.

4.3 Estimation method

Estimation methods refer to how our parameters are calculated. For continuous data, maximum likelihood (or robust maximum likelihood, MLR) is a popular choice. Our data set primarily consists of ordinal variables, which, according to note 8 by Wallentin (2026, Slide 25), if there is enough categories (9+) can be considered for estimation with MLR. As we mostly have ordinals with 5 categories, we opt for a modified weighted least squares estimator: Weighted Least Squares Mean and Variance adjusted (WLSMV). This technique is suitable for working with mixed correlation types.

4.4 Fit Metrics

To test our model fit, we will utilize TLI, CLI, RMSEA and SRMR. The use of the χ^2 generated p-value wasn't used, due to its tendency to reject for large sample sizes (Wallentin, 2026, Slide 14). Reported fit metrics will be the **scaled variants** to correct for polychoric correlations due to ordinal variables.

5 Hypothesis

The process of creating a hypothesized model wasn't completely straightforward. We wanted to stay true to our study purpose of investigating positive and negative perceptions and outcomes of AI-use, which meant ideally having 5 factors (positive/negative perception, AI-use, and positive/negative outcome), but as seen in section 3, correlations weren't completely clear-cut, and we simply did our best with that. With reference to Figure 2 and the literature/frameworks presented in section 1, we build our hypothesis:

In terms of **positive perceptions**, we have three indicators that formed a grouping in our correlation plot that relate to study habits. For us, we believe study habits could be linked with the TAM, where students with worse study habits could perceive AI as more useful, and in turn lead to increased AI-usage. For **negative perceptions**, before observing the correlation plot, we thought *AI use in academic tasks is fraud* along with possibly *perceived dependence on AI* and *perceived creativity reduction due to AI* could be strong together. However, as we saw, the correlations of *fraud* with any other indicator was low, and, the two perception indicators ended up correlating with other indicators better. Thus, we decided not to create a negative perception factor.

An **AI-use** factor was able to be created with the four indicators in the bottom left box of Figure 2; AI-use frequency, dependency, AI helps understand topics, and verify work. They shared good correlations and strong intuitive connections to the latent. An **Academic Success** factor was able to be formed by the second to bottom left box in figure 2. These 4 indicators also had strong correlations, and interpretations for the latent. Finally, a **negative outcomes** factor was created with guidance of Abbas et al. (2024) who had 3 negative factors as outcomes of AI-use; Procrastination, Memory Loss, and Academic Performance. We believe procrastination, creativity reduction, and failing an assignment due to AI were very similar. These are additionally the top green box in Figure 2.

Having formed our four latents, we needed to hypothesise the paths between them. Study habits, which would dictate how useful students perceived AI to be, were given a path to AI-use. AI-use was then given a path to both "Academic Success" and "AI-use risks", as our positive and negative outcome latents. Finally, AI-use risks were also hypothesized to have an effect on academic success. The hypothesized model is visualized in Figure 3.

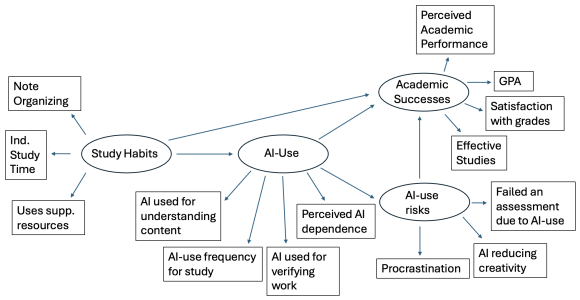


Figure 3: Hypothesis

Finally, we wanted to make some hypotheses on these specified paths between the latents, and we formalize those now. **H0:** Good study habits are inversely associated with AI-usage. **H1:** AI-usage is positively associated with AI-use risks. **H2:** AI-usage is positively associated with academic successes. **H3:** AI-use risks are negatively associated with academic successes. And, **H4:** Study habits are positively associated with academic success.

The model can be expressed in matrix notation as a full SEM model (Wallentin, 2026):

$$\mathbf{x} = \mathbf{\Lambda_x}\boldsymbol{\xi} + \boldsymbol{\delta}$$
$$\mathbf{y} = \mathbf{\Lambda_y}\boldsymbol{\eta} + \boldsymbol{\epsilon}$$
$$\boldsymbol{\eta} = \boldsymbol{B}\boldsymbol{\eta} + \mathbf{\Gamma}\boldsymbol{\xi} + \boldsymbol{\zeta}$$

Where:

$$\mathbf{x} = \begin{bmatrix} x_{\text{Notes}} \\ x_{\text{IndStud}} \\ x_{\text{SuppR}} \end{bmatrix} \quad \mathbf{\Lambda_x} = \begin{bmatrix} \lambda_{\text{SH},1} \\ \lambda_{\text{SH},2} \\ \lambda_{\text{SH},3} \end{bmatrix} \quad \boldsymbol{\xi} = \xi_{SH} \quad \boldsymbol{\eta} = \begin{bmatrix} \eta_{\text{AI}} \\ \eta_{\text{AIR}} \\ \eta_{\text{AS}} \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} y_{\text{AI4Undrstnd}} \\ y_{\text{AIuseFreq}} \\ y_{\text{AI4Verify}} \\ y_{\text{AIDep}} \\ y_{\text{Prcrstinatn}} \\ y_{\text{FailDue2AI}} \\ y_{\text{AICreativity}} \\ y_{\text{SatisfGrdes}} \\ y_{\text{AcadPerf}} \\ y_{\text{GPA}} \\ y_{\text{EffctivStudies}} \end{bmatrix} \quad \mathbf{\Lambda_y} = \begin{bmatrix} \lambda_{\text{AI},1} & & \\ \lambda_{\text{AI},2} & & \\ \lambda_{\text{AI},3} & & \\ \lambda_{\text{AI},4} & & \\ & \lambda_{\text{AIR},1} & \\ & \lambda_{\text{AIR},2} & \\ & \lambda_{\text{AIR},3} & \\ & & \lambda_{\text{AS},1} \\ & & \lambda_{\text{AS},2} \\ & & \lambda_{\text{AS},3} \\ & & \lambda_{\text{AS},4} \end{bmatrix}$$

$$\boldsymbol{B} = \begin{bmatrix} 0 & 0 & 0 \\ \beta_{\text{AI,AIR}} & 0 & 0 \\ \beta_{\text{AS,AI}} & \beta_{\text{AS,AIR}} & 0 \end{bmatrix} \quad \mathbf{\Gamma} = \begin{bmatrix} \gamma_{\text{AI,SH}} \\ 0 \\ \gamma_{\text{AS,SH}} \end{bmatrix}$$

The covariance matrices for $\boldsymbol{\xi}$, $\boldsymbol{\zeta}$, $\boldsymbol{\delta}$, $\boldsymbol{\epsilon}$, are $\mathbf{\Phi}$, $\mathbf{\Psi}$, $\mathbf{\Omega_\delta}$, and $\mathbf{\Omega_\epsilon}$ respectively. For our hypothesis, they are defined as diagonal matrices; $\mathbf{\Phi} = \text{Var}(\xi_{SH})$, $\mathbf{\Psi} = diag\{\text{Var}(\zeta_{AI}), \text{Var}(\zeta_{AIR}), \text{Var}(\xi_{AS})\}$, $\mathbf{\Omega_\delta} = diag\{\text{Var}(\delta_{x_{\text{notes}}}), \text{Var}(\delta_{x_{\text{IndStudy}}}), \text{Var}(\xi_{x_{\text{SuppR}}})\}$, and finally, $\mathbf{\Omega_\epsilon} = diag\{\text{Var}(\epsilon_{y_{\text{AI4Undrstnd}}}), \text{Var}(\epsilon_{y_{\text{AIuseFreq}}}), ..., \text{Var}(\epsilon_{y_{\text{EffctivStudies}}})\}$.

Before moving on, we want to check that our model is identified. We have 14 indicators in our hypothesis. That means 105 pieces of unique information for the model to utilize. Counting the unknown parameters now specified, we have 37. That means we're globally over-identified and have 68 degrees of freedom without using the marker method/variance standardization. Further, we have made sure to put 3 indicators on each latent, satisfying the 3-indicator rule, and can be confident our model is overall identified (see note 4 by Wallentin (2026)).

6 Analysis

6.1 Initial fit

Our analysis begins by fitting the model hypothesized above. Utilizing the `sem` function from the `lavaan` package in R, with the mentioned "WLSMV" estimator and standardized variances (`std.lv = T`), we obtain the following fit measures:

RMSEA	SRMR	ScaledCFI	ScaledTLI
0.098	0.097	0.819	0.771

Table 4: Hypothesised Model Fit Measures

By the benchmarks given in note 4 by Wallentin (2026), our fit is sub-par. Our data does not match our hypothesis. The process of seeing how we can improve our model begins.

6.2 Model Modifications

Starting with our initial fit, we went to the modification indices for suggestions of places where improvements could be made. The top suggestion was to cross-load *Uses Supplementary Resources* onto AI-Use. This change had a modification index, "mi", of 108.798, so it was highly influential.

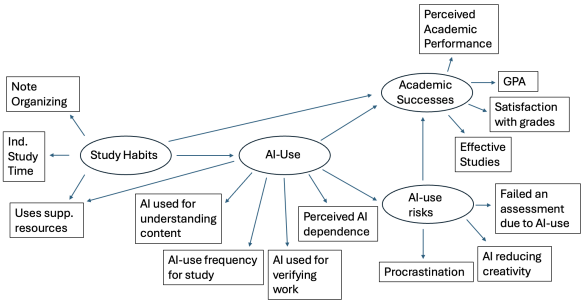


Figure 4: First Model Modification

The suggested change is shown in the path diagram above. It is perhaps not a completely unexpected suggestion from the software. As we saw in the correlation plot (Figure 2), *Uses Supplementary Resources* is closely correlated to the AI-use cluster (see the right-hand side of Figure 2), even more so than to the cluster it ended up being placed in. Contextually, that makes sense. In the survey question for *Uses Supplementary Resources*, it includes AI-use as a suggestion, so it is likely to be correlated with AI-use. Thus, we are happy to make this cross-loading, but we are also now mindful of checking the loadings of *Uses Supplementary Resources* on Study Habits and AI-use afterwards. If it is stronger on the latter, perhaps it is simply better suited to AI-use. The "second fit", as seen in Figure 4, produces the following fit measures:

RMSEA	SRMR	ScaledCFI	ScaledTLI
0.077	0.083	0.890	0.859

Table 5: Second Fit Measures

Which is a vast improvement. The next modification indices suggest more cross-loadings, but this risks diminishing the interpretability of our model, and for now are avoided. Instead, we look at the factor loadings on this fit, specifically at the loadings of *Uses Supplementary Resources* on both Study Habits and AI-use. We find standardized loadings of 0.260 and 0.452, respectively, and consequently decide to remove the indicator from Study Habits.

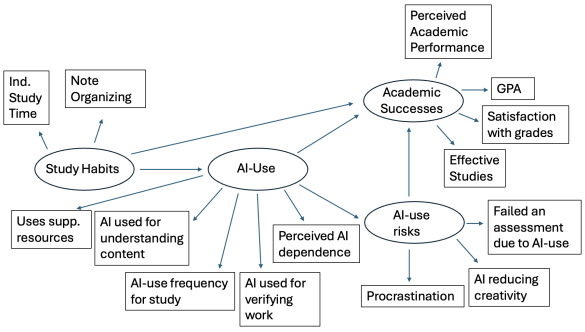


Figure 5: Second Model Modification

The change is visualized above in Figure 5. This model obtains the following fit measures:

RMSEA	SRMR	ScaledCFI	ScaledTLI
0.082	0.086	0.875	0.842

Table 6: Third Fit Measures

As can be seen, the fit measures all worsened, however we're happy to have made the change as we believe it removes what could be considered a controversial factor loading (Quali-Fi, 2026). Continuing on this path of locating weak indicators, we find that the *Failed An Assignment Due to AI* indicator on AI-Use Risks has a weak standardised loading of 0.25 and a resulting R^2 of 6.2%. Our dilemma is that we think this indicator should, in theory, work well for measuring risks of using AI. What better measure of an academic risk of using AI is there than failing an assessment due to it? Therefore, with contextual justification, it was left

in the model.

Looking at modification indices, the top 3 suggestions are to cross-load *Perceived Dependence* onto AI-use Risk or Academic Success, or to allow the errors of *Perceived Dependence* and *AI Reducing Creativity* to correlate. They have 'mi" in the 25-60 range. Although beneficial to our fit measures, we do not want to make these changes. We believe cross-loading or correlating errors of an indicator from AI-Use onto one of our outcome latents blurs the lines of our framework. The outcome latents are interpreted in a way that they are a result of AI-Use, and breaking this boundary with a cross-loading isn't something we want to do.

The next 8 modification indices all lead to a change in "mi" of 24.08 and surround structural edits, whether that be paths or covariances. The only real option without making a recursive model (which we do feel is an interesting bit of research for another time), would be to add a path from Study Habits to AI-Use Risks. However, this does not fit into our framework of those being risks from AI-use. We therefore do not want to take any of those modification suggestions.

One modification was, however, considered. It had a 'mi" of 10.66 and involved correlating the errors between *satisfaction with grades* and *self rated academic performance*. As these two indicators sit on the same latent, we thought it would be acceptable to correlate them. After fitting this model, we found no further modification indices of significant value (mi $>$ 5) that consisted of changes we were willing to make. We reached our final fit.

6.3 Final fit

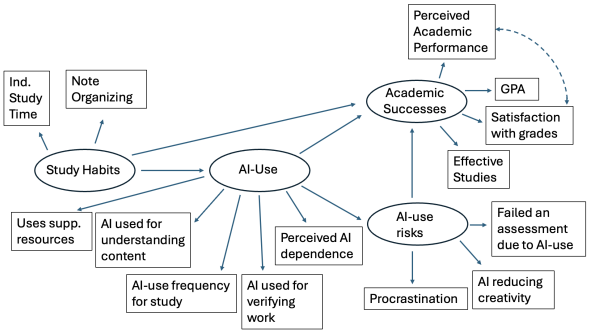


Figure 6: Final Model

A matrix representation of the model can be found in the appendix. The biggest changes from the original hypothesized model is a change of the latent variable for *uses supplementary resources* as well as a correlated error term between *Perceived Academic Performance* and *Satisfaction with Grades*. The model fits for the final model are:

RMSEA	SRMR	ScaledCFI	ScaledTLI
0.080	0.085	0.883	0.850

Table 7: Final Fit Measures

These are all still in the range of barely acceptable to acceptable. We acknowledge that our model does not fit the data very well. We didn't want to make significant modifications that would blur the lines of our research goals. A full discussion of the model comes in the results section.

6.4 Multigroup

The other main research objective was to run a multi-group analysis. However, as we have seen, the final fit wasn't a great fit. Therefore, we were faced with the decision of doing a multi-group on a very average model, and possibly fighting an uphill battle from the start, or simplifying our model to have a more solid starting point. The latter is what we opted for.

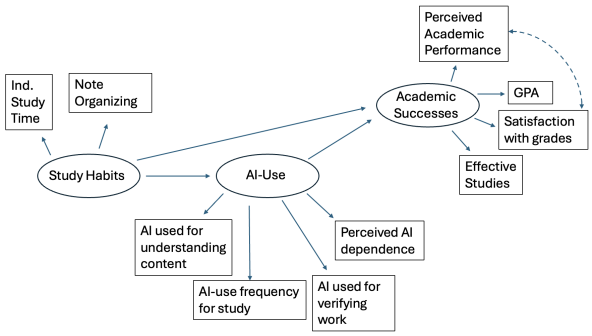


Figure 7: Multigroup model

The model is seen above. The process of selecting it was as follows. **First**, we decided to remove the *Failed an Assessment due to AI* indicator from AI-use risks. With a standardized loading of 0.249, and poor correlations with pretty much every indicator, it didn't need to be there if we ignore it's contextual benefits. **Next**, in the modification indices throughout the main model analysis, it was suggested that AI dependence and AI creativity should be together; we decided to give in to this, and load creativity onto AI-use. **Then**, with one indicator left on AI-Use Risks (Procrastination), we drop the latent entirely - procrastination was in fact not a very strong indicator (in terms of its correlations) in the first place when looking at figure 2. **We run the sem model as we have described up to now** and find Creativity has a standardized loading of 0.356 on AI-Use, and as there are 6 indicators there now, we drop Creativity. **Finally**, we run the model one more time and see that *uses*

supplementary resources and *verifies work with AI* have standardized loadings of 0.552 and 0.462 respectively. We decide to drop *uses supplementary resources* despite its higher loading as we think it has a vaguer contextual interpretation than verifies work with AI. This results in the model seen in Figure 7. The model fits are:

RMSEA	SRMR	ScaledCFI	ScaledTLI
0.067	0.062	0.953	0.932

Table 8: Mulitgroup Model Fit Measures

Which is a good fit per the benchmarks.

We believe the model is still interesting to test with a multi-group analysis. We can still test hypotheses **H0** (if study habits are negatively related to AI-use), **H2** (AI-usage is positively associated with academic success), and **H4** (study habits are positively associated with academic success).

At this point, we make a concession. We were unable to find resources on how to do multi-group analysis with ordinal variables, i.e., comparing thresholds, etc., instead of intercepts. Therefore, we decided to treat our ordinal variables as continuous and use the "Robust Maximum Likelihood" (MLR) estimator. We begin by testing configurate invariance by splitting the data set by gender and running the model on the split data sets. The model fit measures (treating the data as continuous and using MLR) for the full data set, and the grouped data sets, are as follows:

Model	**RMSEA**	**SRMR**	**CFI**	**TLI**
Unsplit	0.046	0.049	0.962	0.945
Women	0.092	0.081	0.863	0.801
Men	0.019	0.050	0.993	0.990

Table 9: Configurate Invariance Test

The overall model fit is now very good when using MLR, which to us indicates the results are possibly untrustworthy/over-ambitious when compared to Table 8. Otherwise, the model shows an acceptable fit for men, with a robust RMSEA and scaled SRMR below 0.08, and a scaled CFI and TLI above 0.9. However, when looking at the model for women, not all of the benchmark values are met. This means the configural invariance is not met, and the specified model does not seem to represent women's responses in the same way it did for men. This means we cannot continue testing for the rest of the measurement and structural invariance tests. However, this is still an interesting result, indicating a difference to the observed relationships, which provided a good fit for men.

7 Results

We present the standardized loadings and regression results from our final fit (section 6.3) in Table 10 on the next page. We see that all 14 indicators chosen have statistically significant loadings onto their respective latent variables. The lowest standardized loading is the discussed *failed evaluated activity due to ai* being just 0.249, but all others are at least 0.42, indicating they are moderately good measurement tools. Interestingly, the strongest indicator for each latent was *organises notes before tests* for study habits (0.769), *AI-use frequency for study* for AI-Use (0.861), *Perceived creativity reduction from AI* for AI-use risks (0.769), and *Study method effectiveness* for Academic Success (0.752).

When looking at the structural path regressions, and with reference to our 5 hypotheses, we found that: **1) H0 was not supported.** Better study habits were found to positively affect AI-Use, not reduce it. That's completely viable, where better study habits is acting in place for something underlying, such as awareness about AI. **2) H1 was supported.** More AI-Usage was found to significantly increase AI-use risks. **3) H2 was not supported.** More AI-use was found to significantly decrease academic success, but it was an insignificant result, indicating maybe academic success comes through other routes. **4) H3 was supported.** AI-use risks/negative outcomes were found to negatively affect academic success. This is logical, if you procrastinate more, fail assignments due to AI, or have a perceived creativity drop, you are likely to perform worse at university. Finally, **5) H4 was supported.** Study habits are positively and significantly associated with academic success.

7.1 Implications within the research

We aimed to develop a model of AI-use that incorporated both negative/positive perceptions and outcomes. That wasn't achieved as hoped, and we wouldn't say this study holds many implications within the research. What we did find was that, in our data, AI-use was indeed associated with negative outcomes, but not directly with academic success. It should be noted that, given the indirect path from AI-use to academic success through AI-use risks, the impact of AI-use on academic success depends on how AI is used and if you are on the receiving end of the negative outcomes, as Hou et al. (2026) proposed. Ultimately, our model fit was on the verge of being acceptable, and therefore, results should be taken with a grain of salt. Our simplified multi-group analysis did show indications of a difference between genders at the configural level, where the underlying structures of how men and women interact with AI within our proposed framework differed in our data.

Latent Constructs & Paths	Standardized Estimate (β / λ)	p-value
Measurement Models		
Study Habits		
organizes_notes_before_tests	0.769	$< .001^*$
independent_study_time	0.494	$< .001^*$
AI-Use		
ai_helped_understand_topics	0.666	$< .001^*$
ai_use_frequency_for_study	0.861	$< .001^*$
verifies_work_with_ai	0.471	$< .001^*$
perceived_ai_dependence	0.728	$< .001^*$
uses_supplementary_resources	0.531	$< .001^*$
AI-Use Risks		
procrastination_frequency	0.420	$< .001^*$
failed_evaluated_activity_due_to_ai	0.249	0.020^*
perceived_creativity_reduction_from_ai	0.769	$< .001^*$
Academic Success		
satisfaction_with_grades	0.556	$< .001^*$
self_rated_academic_performance	0.591	$< .001^*$
cumulative_gpa	0.525	$< .001^*$
study_method_effectiveness	0.752	$< .001^*$
Structural Path Regressions		
AI-Use → Academic Success	-0.084	0.368
Study Habits → Academic Success	0.588	$< .001^*$
AI-Use Risks → Academic Success	-0.219	0.027^*
AI-Use → AI-Use Risks	0.449	$< .001^*$
Study Habits → AI-Use	0.293	$< .001^*$

Table 10: Standardized Factor Loadings, Path Coefficients, and Significance Values

8 Future research

We believe more research into this topic is interesting. Understanding uptake into AI and both the risks and benefits of its use are vital in the AI-filled world we're heading towards. It would be an interesting bit of research to create a questionnaire in line with the mixed perception, mixed outcome framework talked about in this paper. Given more time, our group would try to conduct a study on students at Uppsala University, specifically for the latent constructs discussed, and see the results first-hand. Our multi-group investigation showed further research into gender differences in relation to AI-use in academia could also be worthwhile.

9 Conclusion and reflection

The results from the CFA confirmed that all variables chosen for the respective latent factors, study habits, AI-use, AI-risks, and academic successes were relevant indicators, all showing significant p-values ($p < 0.001$). However, since the fit measures showed to be barely acceptable, conclusions to this model should be drawn with caution.

The SEM model revealed some interesting relationships. Study habits had a significant positive estimate for academic success (0.293), a relationship that went against the initial hypothesis. Study habits significantly increased academic success (0.588), a relationship that very much makes sense. Increased AI use was found to increase AI-use risks (0.293). Finally AI-use significantly increased AI-use risk, which in turn had a negative impact on academic performance. However, just as with the CFA model, because of the fit measures, these also need to be interpreted with caution, with further research needed to verify these relationships.

Finally, the multi-group model failed the configurate invariance test. The testing done for the simpler multi-group model found that men showed a drastically better fit, compared to the model for women that gave metrics that were below benchmarks. This indicates that there seems to be a difference in these relationships between men and women, something that would be interesting to study in further research.

As a group, we absolutely faced some challenges along the way to completion. **For one,** finding a publicly available data set that was suitable for our research objectives. We couldn't find a data set that would permit us to test our complete intended hypothesis, but we're still pleased with the research process and thankful to Correa (2026) for

making his data publicly available. As mentioned, we would love to run a survey on students at Uppsala University in this subject. **We also feel** we got a great experience of the decision making process of making modifications to a model. It can be tempting to go model fit measure chasing but it has to be done within the constrains of your framework. We felt we showed good discipline on this front. **Finally,** it was a good learning opportunity to again see that research doesn't always go according to plan. Changing the final model to a 3-factor SEM to do the multi-group analysis wasn't initially planned but we learnt you have to be ready to adjust to the results in front of you.

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10 Appendix

Academic Program	Enrolled 2025_1	Planned # For Sample	Records in Survey
Business Administration	838	60	60
Information Systems Mngment	412	30	30
Architecture	540	39	39
Computer Science	189	14	14
Cultural & Communication Mngment	116	8	8
Civil Engineering	509	37	37
Electrical Engineering	460	33	33
Electronic Engineering	392	28	28
Engineering Physics	272	20	20
Industrial Engineering	511	37	37
Chemical Engineering	476	34	34
Biological Engineering	83	6	6
Mathematics	115	8	8
Statistics	37	3	3
Total	4950	357	357

Table 11: Distribution of enrolled students and students in the survey

Final Model Matrix Notation:

$$\begin{matrix}\mathbf{x} = \mathbf{\Lambda_x}\boldsymbol{\xi} + \boldsymbol{\delta} \\ \mathbf{y} = \mathbf{\Lambda_y}\boldsymbol{\eta} + \boldsymbol{\epsilon} \\ \boldsymbol{\eta} = \boldsymbol{B}\boldsymbol{\eta} + \mathbf{\Gamma}\boldsymbol{\xi} + \boldsymbol{\zeta}\end{matrix} \quad \text{Where: } \mathbf{x} = \begin{bmatrix} x_{\text{Notes}} \\ x_{\text{IndStud}} \end{bmatrix}, \mathbf{\Lambda_x} = \begin{bmatrix} \lambda_{\text{SH},1} \\ \lambda_{\text{SH},2} \end{bmatrix}, \boldsymbol{\xi} = \xi_{SH}, \boldsymbol{B} = \begin{bmatrix} 0 & 0 & 0 \\ \beta_{\text{AI,AIR}} & 0 & 0 \\ \beta_{\text{AS,AI}} & \beta_{\text{AS,AIR}} & 0 \end{bmatrix},$$

$$\mathbf{\Gamma} = \begin{bmatrix} \gamma_{\text{AI,SH}} \\ 0 \\ \gamma_{\text{AS,SH}} \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_{\text{SuppR}} \\ y_{\text{AI4Undrstnd}} \\ y_{\text{AIuseFreq}} \\ y_{\text{AI4Verify}} \\ y_{\text{AIDep}} \\ y_{\text{Prcrstinatn}} \\ y_{\text{FailDue2AI}} \\ y_{\text{AICreativity}} \\ y_{\text{SatisfGrdes}} \\ y_{\text{AcadPerf}} \\ y_{\text{GPA}} \\ y_{\text{EffctivStudies}} \end{bmatrix}, \quad \boldsymbol{\eta} = \begin{bmatrix} \eta_{\text{AI}} \\ \eta_{\text{AIR}} \\ \eta_{\text{AS}} \end{bmatrix}, \quad \mathbf{\Lambda_y} = \begin{bmatrix} \lambda_{\text{AI},1} & & \\ \lambda_{\text{AI},2} & & \\ \lambda_{\text{AI},3} & & \\ \lambda_{\text{AI},4} & & \\ \lambda_{\text{AI},5} & & \\ & \lambda_{\text{AIR},1} & \\ & \lambda_{\text{AIR},2} & \\ & \lambda_{\text{AIR},3} & \\ & & \lambda_{\text{AS},1} \\ & & \lambda_{\text{AS},2} \\ & & \lambda_{\text{AS},3} \\ & & \lambda_{\text{AS},4} \end{bmatrix}$$

$\mathbf{\Phi} = \text{Var}(\xi_{SH})$, $\mathbf{\Psi} = diag\{\text{Var}(\zeta_{AI}), \text{Var}(\zeta_{AIR}), \text{Var}(\xi_{AS})\}$, $\mathbf{\Omega_\delta} = diag\{\text{Var}(\delta_{x_{\text{notes}}}), \text{Var}(\delta_{x_{\text{IndStudy}}}), \text{Var}(\xi_{x_{\text{SuppR}}})\}$, and finally, $\mathbf{\Omega_\epsilon} = diag\{\text{Var}(\epsilon_{y_{\text{AI4Undrstnd}}}), \text{Var}(\epsilon_{y_{\text{AIuseFreq}}}), ..., \text{Var}(\epsilon_{y_{\text{EffctivStudies}}})\}$ **and a covariance** $\text{Cov}(\epsilon_{y_{\text{SatisfGrdes}}}, \epsilon_{y_{\text{AcadPerf}}})$